



gain
Global Alliance for
Improved Nutrition

Training Manual

Basic and Advanced Food Safety and Hygiene, Best Practices and Regulatory Standards

Strengthening Nutrition in Priority Staples

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About GAIN

The Global Alliance for Improved Nutrition (GAIN) is a Swiss-based foundation launched at the UN in 2002 to tackle the human suffering caused by malnutrition. Working with both governments and businesses, we aim to transform food systems so that they deliver more nutritious food for all people. At GAIN, we believe that everyone in the world should have access to nutritious and safe food. We work to understand and deliver specific solutions to the daily challenge of food insecurity faced by poor people. By understanding that there is no “one-size-fits-all” model, we develop alliances and build tailored programmes using a variety of flexible models and approaches. We build alliances between governments, local and global businesses, and civil society to deliver sustainable improvements at scale.

We are part of a global network of partners working together to create sustainable solutions to malnutrition. Through alliances, we provide technical, financial and policy support to key participants in the food system. We use specific learning, evidence of impact, and results of projects and programmes to shape and influence the actions of others. Headquartered in Geneva, Switzerland, GAIN has representative offices in Denmark, The Netherlands, the United Kingdom, and the United States. In addition, we have country offices in Bangladesh, Ethiopia, India, Indonesia, Kenya, Mozambique, Nigeria, Pakistan, and Tanzania. Programmes and projects are carried out in a variety of other countries, particularly in Africa and Asia.

GAIN supports businesses, especially micro, small, and medium-sized enterprises (MSMEs), through most of our programmatic offerings. This support is coordinated through the Scaling Up Nutrition (SUN) Business Network (SBN), which is convened by GAIN in Nigeria and was launched in 2016 to improve and strengthen the private sector’s contribution to nutrition. The Network has supported businesses with brokering business-to-business linkages, facilitating connections to Government and other sectors, including development partners, and capacity building for MSMEs to address critical barriers to start, grow and scale their business. Specific support areas include access to finance, access to markets, quality control and assurance in their production processes, and food safety and nutrition education.

Project Background

The Strengthening Nutrition in Priority Staples project is currently implemented by The Global Alliance for Improved Nutrition in partnership with GIZ and the Green Innovation Centre for the Agricultural and Food Sector in Nigeria (GIC), with support from the German Government through the Federal Ministry for Economic Cooperation and Development (BMZ). The Strengthening nutrition project will strengthen the GIC priority value chains and improve food security and safety through improved diets for farmers, farm workers, other workers in businesses in these value chains, and among the wider population in Nigeria.

The business support component will empower women and young people and unleash the creativity and energy of youth and women to support their families, succeed in agri-business and attempt to tackle malnutrition and prevent foodborne illnesses as well as food and social insecurity. This workstream will increase women and youth involvement in value-added agriculture and entrepreneurship that directly addresses the specific challenges women and rural youth face in the four focal states for this project. It will respond to opportunities to harness the political will at local, state, and national levels in Nigeria to invest in creating economic opportunities for women and youth in the production, processing, and sale of nutritious and safe foods.

This will be achieved by significantly increasing the knowledge and information available to women and young people about nutrition, and food safety, increasing their participation in decision-making and targeting business support and investments so they can bring forward innovations, increase livelihood opportunities and build nutrition sensitivity into the GIC value chains.

This will be carried out using three interlinked approaches:

- Improving the nutrition quality of staple foods produced by SMEs and farmers involved in GIC activities.
- Improving the capacity of food safety and value chain actors to adhere to standards and regulations on best practices.
- Supporting technical innovations for supply chains through innovation challenges and innovation clinics

Rationale

Ensuring food and nutrition security is one of the most pressing challenges for sustainable social and economic development in Nigeria. Meeting this need presents an important opportunity for tackling malnutrition. Diets of rural Nigerians are particularly poor, and many Nigerians currently experience vitamin and mineral deficiencies leading to severe health outcomes including premature and preventable death especially in under 5-year olds.

According to NDHS 2018, approximately 66% of Nigerian children are at risk of vitamin A deficiency due to a lack of diversity in their diets as a consequence of poverty. This “hidden hunger” is particularly prevalent among farming families who are overly reliant on staple foods. The number of people at risk from unsafe food is also alarming - an estimated 173 million cases of diarrhea due to foodborne illness and about 33,000 deaths are reported in Nigeria. Food safety is also a problem in informal markets where these SMEs operate. Poor food safety regulations/ policies, poor food safety knowledge and practices mean that consumers are exposed to unsafe foods. SMEs, who handle most of the food that is produced, transported, marketed, and consumed in low-income countries such as Nigeria are hindered by a lack of business knowledge and skills, and often lack resources for innovation.

GAIN will leverage on its existing work and partnerships in the selected implementation states to achieve the set objectives of this Project, ultimately impacting positively on the nutrition indices of Nigerians who live and work in these states and beyond.

Training Objective

To enhance the technical skills of food processors to meet basic and advanced food safety and hygiene, best practices and regulatory standards.

Learning Outcomes

At the end of the training, participants should be able to:

- Know the different types of food safety hazards.
- Identify the possible sources of contamination of food.
- Understand the importance of food safety in processing operations.
- Understand the importance of quality control and assurance to ensure safe and quality products.
- Improve on basic hygiene practices in processing operations.
- Promote Good Manufacturing Practices during processing operations.
- Understand the regulatory guidelines on safe and quality products.

Module 1: Principles Of Food Safety And Hygiene

Introduction

- Consumers expect the food they eat to be safe, however people in every country both developed and developing still fall ill from the food they eat.
- Safe and nutritious food is necessary for human survival and well-being.
- Food safety involves protecting food from anything that can cause harm or illness to the consumer.
- With more people travelling, and food supply chains becoming increasingly global, foodborne diseases, like other diseases, have the potential to spread widely.
- Foodborne illnesses are caused by dangerous micro-organisms and/or toxic chemicals e.g., bacteria, viruses, parasites, moulds, chemicals & metals
- Food safety is a shared responsibility between governments, producers, and consumers.
- Everybody has a role to play from farm to table to ensure that the food we consume is safe and will not cause any harm to our health.
- Farmers, processors, transporters, and distributors of food must obey food hygiene rules to prevent food borne illnesses.
- Food businesses should know the hazards that may affect their products and the consequences of these hazards for consumer health and ensure that they are prevented.
- Good Hygiene Practices (GHPs) are the foundation of any effective control of hazards associated with food.
- Effective implementation of GAP & GHP should be sufficient to address food safety issues at the farm level, while processors should implement GHP, GMP and HACCP to ensure that control measures are effective.
- Food borne illness should be prevented because it is a burden to the healthcare system, It weakens the economy and national development efforts, as well as international trade.

Food Safety Terms & Definitions

Contaminant: Any biological, chemical, or foreign matter or other substances not intentionally added to food that may compromise food safety or suitability.

Contamination: the presence or introduction of a contaminant in the food or food environment.

Cross-contamination: The transfer of contaminants particularly bacteria from contaminated foods, generally raw to ready-to-eat foods by direct contact, drip or indirect contact using a vehicle such as hand, clothes, knife, thermometers, etc.

Control Measure: Any action or activity that can be used to prevent or eliminate a hazard or reduce it to an acceptable level.

Food hygiene: All conditions and measures necessary to ensure the safety and suitability of food at all stages of the food chain

Food safety: The assurance that food will not cause harm to the consumer when it is prepared and/or eaten according to its intended use

Food quality: The sum of all properties and attributes of a food item that are acceptable to the customer.

Hazard: A physical, chemical, or biological agent or substance with the potential to cause an adverse health effect.

Prerequisite programme: Programmes including Good Hygiene Practices, Good Agricultural Practices and Good Manufacturing Practices, as well as other practices and procedures such as training and traceability, that establish the basic environmental and operating conditions that set the foundation for implementation of a HACCP system.

Food Safety

- Food safety involves producing, handling, storing, and preparing food in a manner that it is not contaminated hence preventing food borne illness.
- Safe food protects consumers from food borne illnesses, increases consumer confidence in the food supply system, improves the earnings of operators, encourages tourism, and enhances national and international food trade which boosts the economy of a nation.
- The current approach to food safety depends on the development of control measures and monitoring such measures, based on analysis of the risks connected to the production processes employed by the producer namely:
 - i. the company's own realities,
 - ii. resources,
 - iii. equipment,
 - iv. staff qualifications,
 - v. regulatory requirements for its products and
 - vi. customers' requirements etc.
- The system focuses on risk analysis and producers are responsible for organizing efficient safety and quality management systems, based on HACCP and self-evaluation of their practices.

Food Safety Hazards

The 3 types of food safety hazards are physical (objects), chemical (poisons) and biological (germs) hazards.

Hazards arise from:

- Improper agricultural practices.
- Polluted environment (e.g., mercury, cadmium, nickel).
- Poor hygiene at any stage of the food chain (e.g., E. coli, Listeria, Campylobacter).

- Lack of preventive controls in food processing and preparation operations.
- Untrained personnel.
- Misuse of chemicals (e.g., pesticides, weed inhibitors, growth hormones).
- Use of banned dyes and flavours (e.g., Sudan red, tartrazine, azo dyes, butter yellow).
- Contaminated raw materials, ingredients, and water (e.g., bacteria, viruses, mycotoxins).
- Inadequate or improper storage or handling.
- Adulterated food and food fraud (intentional addition of cheaper varieties, substitution with alternate substances etc.)

Possible Sources of Food Safety Hazards

Physical Hazards



Products and Packaging	Insect	
	String	
	Soft Plastic	
	Cardboard	
	Paper	
	Staples	
	Sticky Paper	
	Sticky Tape	

Premise	Glass	
	Perspex	
	Metal	
	Wood	
	Frayed Belts	
	Flaking Paint	
	Nuts and Bolts	
People	Hair	
	Jewelry	
	Hair Nets	
	Clothing Fibres	
	Gloves	
	Pens	

Chemical Hazards

- Chemicals from non-food grade packaging in contact with food can also leach into food.
- Substances like heavy metals (lead, cadmium, zinc, cyanide), refrigerants etc. may be present in food due to pollution.
- Detergents and lubricants used by the food handlers or processors may also contaminate food and present a hazard.
- Chemical hazards exist also in natural form such as allergens, mycotoxins, alkaloids (glycoalkaloids in potatoes), enzyme inhibitors.

Mycotoxins are:

- Poisonous chemical compounds produced by certain fungi associated with diseased or mouldy crops.
- They can be found in cassava, seeds & nuts such as peanuts; grains such as rice, wheat, maize, barley; and fruits such as apples

Allergens are:

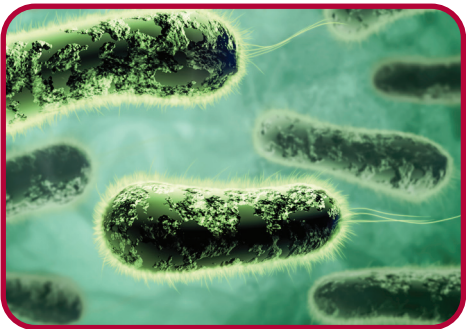
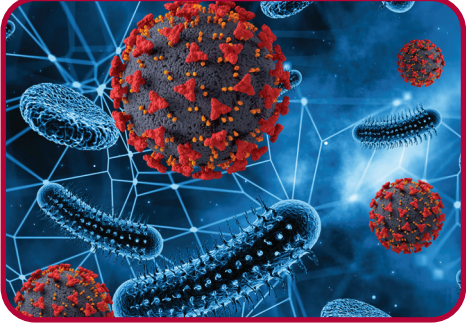
- Proteins in foods or derivatives of them that cause abnormal immune responses.
- Examples of food that contain allergens are wheat, milk., eggs., peanuts., soy., tree nuts (such as walnuts and cashews), shellfish (such as shrimp)

Naturally Occurring	Aflatoxins	
	Mycotoxins	
	Ochratoxins	
	Scombrototoxin (histamine)	
	Ciguatoxin (fish)	
	Fungi toxins (mushrooms)	
Intentionally added	Agricultural chemicals – animal health treatments, pesticides, herbicides, insecticides	
	Preservatives in excess of permitted levels - SO2	
	Processing aids in excess of permitted levels – potassium sorbate, Propionic acid	
Contaminates	Environmental vs. processing	
	Plasticizers	
	Printing inks	
	Adhesives	
	Equipment lubricants	
	Cleaning and sanitising products	
	Paint and flooring materials	
	Water and steam treatment chemicals	
	Pest control pesticides and insecticides	
	Refrigerants	

Biological Hazards

- Biological hazards are harmful micro-organisms such as bacteria, viruses, helminths, protozoa, algae, and certain toxic products they may produce.
- Not all micro-organisms are harmful.
- Bacteria in yogurt, yeast in bread, or certain molds in cheese form part of the food production and play an important role to impart characteristic flavours.
- Biological hazards are introduced into food primarily through:

- contact with soil,
- contaminated water,
- incorrectly treated manure,
- sewage,
- air,
- persons,
- animals (farm animals, birds, and pests), and
- transport.

Bacteria spore forming	Clostridium botulium	
	Clostridium pefringens	
	Bacillus cereus	
Bacteria non spore forming	Campylobacter spp.	
	Shiga toxin-producing E. Coli	
	Listeria mono.	
	Salmonella spp.	
	Shigella spp.	
	Staphylococcus aureus	
	Vibrio spp.	
	Yersinia enterocolitica	
Viruses	Hepatitis A and E	
	Norwalk viruses	
	Rotaviruses	

Protozoa and parasites	Cryptosporidium parvum	
	Trichinella spiralis	
	Cyclospora	
	Cayetanensis	
	Toxoplasma gondii	

General Principles of Food Hygiene

- Food safety and suitability should be controlled using a science-based preventive approach such as a food hygiene system. GHPs ensure that food is produced and handled in an environment that minimizes the presence of contaminants.
- Properly applied prerequisite programmes, which include GHPs, provide the foundation for an effective HACCP system.
- Each food business operator should be aware of the hazards associated with the raw materials and other ingredients, the production or preparation process, and the environment in which the food is produced and/or handled.
- Depending on the nature of the food, food process, and the potential for adverse health effects, it may be sufficient to apply GHPs to control hazards.
- When the application of GHPs alone is not sufficient, a combination of GHPs and additional control measures at Critical Control Points (CCPs) should be applied.
- Control measures should be subject to monitoring, corrective actions, verification, and documentation, as appropriate to the nature of the food product and the size of the food business.
- Food hygiene systems should be reviewed periodically and whenever there is a significant change that could impact the potential hazards and/or the control measures (e.g., new process, new ingredient, new product, new equipment, new scientific knowledge) associated with the food business.
- Appropriate communication about the food and food process should be maintained among all relevant parties to ensure food safety and suitability across the entire food chain.

Management Commitment to Food Safety

A positive food safety culture which recognizes the importance of human behaviour in providing safe and suitable food is important for the successful functioning of any food hygiene system.

- The following elements are important in cultivating a positive food safety culture:
- Commitment of the management and all personnel to the production and handling of safe food.
- Leadership to set the right direction and to engage all personnel in food safety practices.

- Awareness of the importance of food hygiene by all personnel in the food business.
- Open and clear communication among all personnel in the food business, including communication of deviations and expectations; and
- The availability of sufficient resources to ensure the effective functioning of the food hygiene system.
- Management should ensure the effectiveness of the food hygiene systems by:
- Clearly communicating roles, responsibilities, and authorities to personnel in the food business.
- Maintaining the integrity of the food hygiene system when changes are planned and implemented.
- Verifying that controls are carried out and working and that documentation is up to date.
- Ensuring that the appropriate training and supervision are in place for personnel.
- Ensuring compliance with relevant regulatory requirements; and
- Encouraging continual improvement, where appropriate, considering developments in science, technology, and best practice.

Regulatory Requirements

- Regulatory requirements cover the general hygiene principles which basically concern the following:
- Hygiene measures related to production conditions (healthy operating premises and packing station).
- Measures for personnel hygiene (health status, personal cleanliness, clothing, access to facilities, etc.).
- Aspects related to control of operations (raw materials, water quality, etc.).
- Aspects related to the facility, maintenance, cleaning, and waste management.
- Aspects related to handling, transport, and storage of products.

Production Conditions from Production to Processing Facility

- Production areas (fields, orchards, nurseries, etc.) and packing and storage areas (sheds, silos, cold storage rooms, etc.) must be kept clean.
- Sanitary measures should be in place to prevent contamination from the soil or in sorting, grading, and packing equipment.
- Waste should be removed regularly.
- Locate areas for waste and compost away from production and packaging areas.
- Provide sanitary facilities in fields and in packaging stations to prevent contamination from faecal matter.
- Toilets, compost, and heaps of manure should not be located near a source of irrigation water or in an area that may flood during heavy rainfalls.
- Plant protection products and fertilizers must be stored in safety and well managed to prevent unauthorized access, leaks, and spills.
- Water must be available for workers to wash their hands.

Personnel

- Personnel who come directly or indirectly into contact with food should be trained, and/or instructed in food hygiene to a level appropriate to the operations they are to perform.
- All personnel (team leaders, full-time and part-time employees as well as seasonal workers) must be aware of their role and responsibility in protecting food from contamination or spoilage.
- All producers, processors and employees must have **practical knowledge of basic sanitation rules and hygiene measures related to the work they perform and the responsibilities they hold**.
- Those who handle cleaning chemicals or other potentially hazardous chemicals (e.g.: plant protection products) should be instructed on how to handle them without risk to themselves (Good Hygienic Practices), the environment (Good Plant Protection Practices) or food products & environment (Good Agricultural Practices).
- Training programmes must be based on a needs analysis, validated by the company's management, regularly reviewed, and updated where necessary.

Water Quality

- Water is used for irrigation, the preparation of mixtures for plant protection treatment and the application of fertilizers in fields.
- It is also used to clean not only fruit and vegetables but also sorting tables, floors and harvesting containers, and so on.
- Protect water sources (ground water, catchment areas).
- Avoid polluting water during growing and packaging operations (discharges).
- Overuse of nitrate fertilizers leads to excessive concentrations of nitrates in ground water.
- Avoid allowing water contaminated or polluted by pathogens or chemical products to come into contact with products.
- When preparing mixtures, avoid contaminating water points with plant protection products.
- Only potable water should be used for food processing (when water is involved in the process).
- Regular water quality analyses are essential.

Facilities

- Locate processing facility far from polluted areas, flood plains or industrial areas to reduce contamination.
- Premises, equipment, and facilities must meet the following characteristics:
 - i. Workflow from dirty to clean areas to keep finished products from meeting raw materials.

- ii. Finished products must be separated from raw materials in the storage area.
- iii. Design and layout of facilities and equipment for sorting, grading, packaging, and processing fresh products must accommodate suitable maintenance, cleaning, and disinfection.
- iv. Surfaces of walls, floors, doors, and ceilings must be smooth and impermeable, allow easy drainage, and be easy to clean and maintain and food contact surfaces must be non-toxic.
- v. It should be possible to control temperature, humidity, air circulation, etc. where necessary
- vi. Effective protection to prevent pest access and harbourage.
- vii. Windows must be easy to clean, and if necessary, fitted with removable and cleanable screens to keep out insects.

Transport & Storage Hygiene

- Special attention should be given to product transport conditions to protect food from potential sources of contamination.
- Loading platforms and containers must be cleaned.
- Products must be protected against damage (cover lorries with a tarpaulin; package products correctly; protect them from the sun and rain, dust, smoke, birds, and other animals; avoid bulk transport which can crush products; avoid the use of used bags; etc.).
- The same vehicle should not be used to transport fruit and vegetables, fertilizer, and plant protection products, or even people!

Module 2: Food Safety In Processing Operations

Food Processing

- Food processing is the conversion of agricultural products into consumable products with desired texture, sensory and nutritional properties using commercially viable methods.
- Primary processing involves cutting, cleaning, packaging, storage, and refrigeration of raw foods to ensure that they are not spoilt before they reach the consumer.
- Food processing methods include drying, smoking, frying, homogenization, pasteurization, and canning.



Cassava



Garri



Cassava Flour



Cassava Chips



Cassava Starch



Maize & Maize Products

Control of Biological Hazard

- Food must not be processed in an unsanitary condition/environment.
- Water used in food preparation/processing must be properly treated with disinfectants. Water analysis must be carried out regularly to ensure it is safe and has no chemical residue such as chlorine.
- Food must be properly cooked/heated to kill pathogenic organisms.
- Food processing operations such as pasteurization/sterilization should be adequate to destroy micro-organisms that may be present in food.
- Moisture content, pH and acidity should be analyzed in food products to ensure food products do not provide a media for microbial growth.
- Food must be stored under specified conditions such as freezing and refrigeration.
- Microbiological analysis must be carried out to ensure food is safe and microbial load is within acceptable limits.
- Food handlers must maintain absolute personal hygiene and food handler test done at least annually. Sick personnel should be granted leave to visit the hospital for treatment.
- Waste must be properly handled and disposed.
- Pest control must be adequate to prevent the entry of pests to contaminate food products leading to the growth of the microorganisms.
- Cleaning of work surface must be done daily. Equipment must also be cleaned using Cleaning in Place (CIP) or Cleaning Out of Place (COP).

Control of Physical Hazard

- Raw materials to be used in food processing must be analyzed for foreign materials contamination to ensure safe food product for consumption.
- Maintenance must be carried out adequately in processing facilities on machines such as destoners, sieves, filters, graders, separators, colour sorting machines and detectors to prevent entry of physical hazards into food products. Calibration must also be carried out regularly on scales used in analysis.
- Personnel in food plants must avoid the use of jewellery and other materials that may fall into food products.
- Use of glass thermometer and other glass wares should be avoided in production areas/shop floor.
- Torn gloves/apron, and hairnet should be discarded, and personnel must be careful to avoid entry into food products during processing. Personnel must wear hairnet always with beards covered with a beard snood or mask.
- Slip sheets should be used between pallets and food products/raw materials during storage in the warehouse.

Control Of Chemical Hazard

- Maximum pesticide residue limits must be adhered to in all food products
- Materials Safety Data Sheet (MSDS) which includes information about potential risks and hazards to workers handling toxic substances must be kept and used in all food plants to regulate use to avoid contamination, cross-contamination, and injury to personnel.

- Pest control should be outsourced to authorized consultants who will carry out fumigation and issue Pest Control Certificates to enable food businesses to meet up with regulatory requirements. If the company has the capacity, only trained personnel should carry out fumigation.
- All raw materials used for production must be accompanied by a Certificate of Analysis (COA) or analysed in the laboratory to avoid processing foods contamination with chemical hazards.
- Pest control plan and records must be kept in all food plants.
- Metals used in the fabrication of processing equipment; facilities/packages must be food grade to avoid metallic contamination. Processors must understand the interaction of food products and packaging materials in the selection of packages.
- Industrial waste must be treated before release to the environment to prevent contamination of farmland for agriculture and water used in food preparation/processing.
- Standards set by regulatory agencies such as the National Environmental Standards and Regulations Enforcement Agency (NESREA) must be strictly implemented to avoid violation.

Prevent Contamination from Employees

- Any behavior or unhygienic practices that could result in food contamination is not allowed in food handling areas e.g., eating, chewing gum, spitting.
- Health status of the individual must be determined before employment to work in food handling areas.
- Employees should have adequate knowledge of the required hygienic practices acquired through training, coaching, work instructions and experience.

Prevent Contamination from Equipment

- Equipment design should permit adequate cleaning, disinfecting and maintenance.
- Equipment and containers should be made of materials with no toxic effect on intended use.
- Equipment should be durable and made of stainless steel preferably.

Prevent Contamination from Storage of Non-food Items

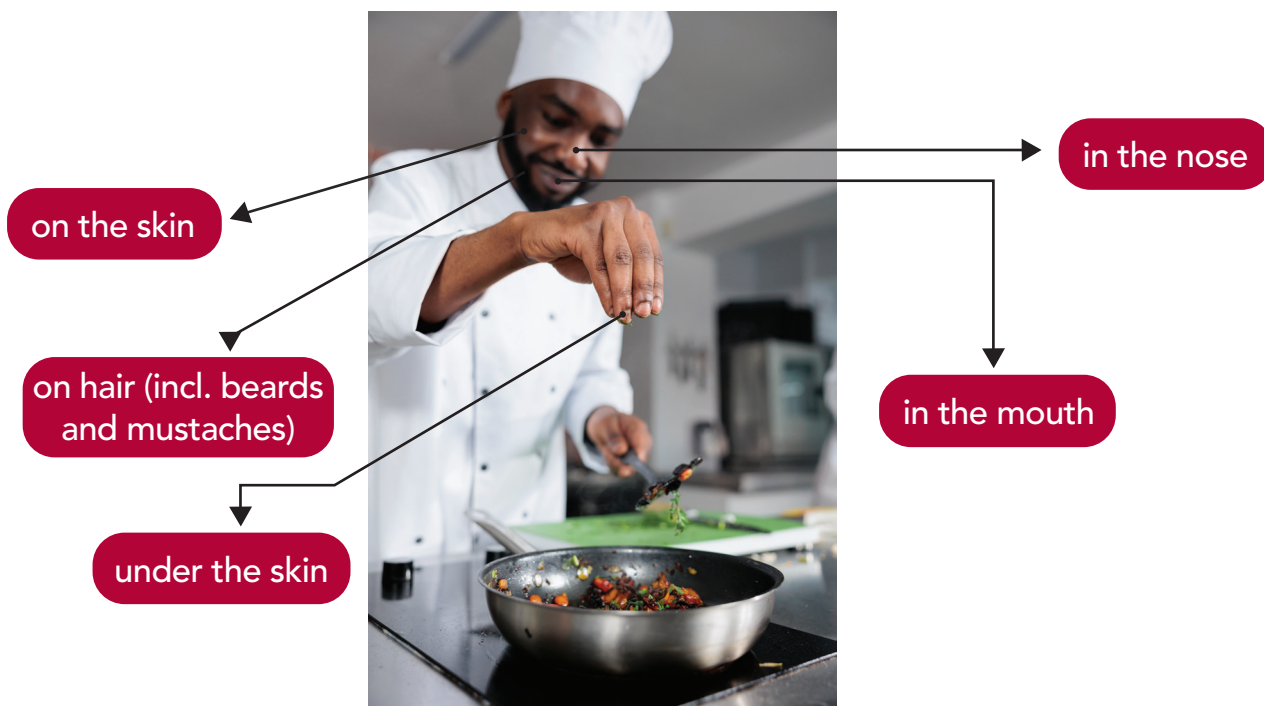
- Handle non-food items carefully to prevent cross-contamination.
- Chemicals should be received and stored in a dry, well-ventilated area, separate from all food handling areas.
- Store chemicals in designated areas to prevent cross contamination and in a manner that prevents contamination of food, food contact surfaces, or packaging materials.
- Chemicals must be stored and mixed in clean and correctly labelled containers.
- Chemicals must be dispensed and handled only by authorized and properly trained personnel

Module 3: Food Handling Practices

Personal Hygiene

- People are a common source of pathogenic bacteria.
- The best approach to avoid contamination is to prevent it.
- All employees working in direct contact with food, food contact surfaces and food packaging must conform to hygienic practices to protect food against contamination by microorganisms or unwanted material.
- Poor hygienic practices and careless employee behaviour can lead to product contamination.
- Provide written instructions on personal hygiene to workers & display it in prominent locations.
- Train workers in personal hygiene practices and keep training records.

A healthy human has microorganism on/in part of the body



Hygiene & Communicable Disease

Areas of focus include clothing, hand washing, personal habits, illness and disease, injuries, visitors, and training.

Production Employees

Employees involved in production should:

- Take a bath daily to be clean when they resume work.
- Wear clean clothes.
- Not use perfume, aftershave, fragrant creams that will affect the product.
- Not wear jewellery that could accidentally fall into the product.

- Not wear false nails or nail polish.
- Keep fingernails short.
- Not eat, drink, or chew gum in the production areas.
- Wash hands thoroughly after using the toilet.
- Not eat over unprotected food or surfaces that are likely to come into contact with food.
- Not sneeze, blow nose or cough over unprotected food or surfaces that are likely to come in contact with food.
- Not spit, smoke, or use tobacco or similar preparations in food production area.
- Lookout for hazards and remove any that they find.

Clothing

- Wear suitable protective clothing (hair net, beard net, footwear, apron, etc) to prevent product contamination.
- Wear separate shoes in the factory.
- Close toes shoes should be worn (no open toes or high heel shoes).
- Personal belongings and street clothing must be stored in locker rooms.
- Do not wear protective clothing outside the production area or hang in sanitary rooms.
- Wear hair and beard nets.
- Wear clean and sanitary gloves when used.



Illness

Provide doctor's certificate of medical screening otherwise known as Food Handlers' Test. The screening should include the following:

- Urinalysis
- Widal test (typhoid fever test).
- Hepatitis B
- Sputum test
- Stool screening

Tell your supervisors if you are ill with symptoms that could contaminate ingredients or products.

- Do not handle food if you are suffering from diarrhoea, sore throat, vomiting, fever, jaundice, any skin wound or discharges from your eyes, nose and ears to prevent product contamination.
- No medication is allowed in the factory.
- Cover any open wound with a clean bandage.

Hand Washing

All employees must wash their hands thoroughly:

- When they enter food handling areas.
- Before starting work.
- After handling contaminated materials.
- After breaks.
- After using toilet facilities.
- As often as is necessary to prevent product contamination.

STANDARD PROCEDURE FOR HAND WASHING



Personnel

- Do not leave gloves, masks, etc. lying around while on break or at shift end.
- Crates boxes, containers or buckets must not be placed directly on the floor.
- Store brooms and dust pans at stations provided.
- Do not lean, sit, or step on product surfaces.
- Do not handle ingredients or products with either cut or infected hand.

Product

- Keep hand contact with ingredients and product to a minimum.
- Check ingredients for expiration dates to ensure that fresh ingredients are used.
- Always keep cooling products covered.

Packaging

- Packaging design & materials should provide adequate protection, prevent damage & accommodate proper labelling.
- Materials or gases must be non-toxic
- Primary packaging material must be food grade.
- Reusable packaging should be durable, easy to clean and disinfect where necessary.

Equipment, Containers & Utensils

Ensure that all containers, including those holding rework, are properly labelled and are kept covered



Use different colour (colour-coded) containers to store ingredients and rework



Use color-coded covered containers for garbage



Use white - or light-coloured cloths to wipe hands regularly and dispose of soiled cloths immediately



Use coloured cloths to clean the floor and objects (e.g., step stools) that come into contact with the floor

- Provide adequate product information to enable the next person in the food chain handle, display, store & use product safely.
- Indicate Lot number, production date and best before date on product label.
- Provide product labels with clear instructions to inform, handle, display, store & use safely.

Sanitation

- Production equipment/utensils must be thoroughly cleaned and sanitized after use.
- Keep contact surfaces clean and free of contamination from tools, cords, cleaning utensils, machine parts, lubricants, and paper.
- Clean all spills promptly.
- Keep everything off the floor and the area clean and floors swept.
- Clean work areas regularly throughout the shift.
- Keep your immediate working area swept or dust mopped. Wipe or mop up spilled liquids promptly.
- Scrape the floor around the work area after completing a job.
- Scrapers for moulds and tabletops are not to be used on the floor.
- Leave your work area clean at the end of your shift.

Receiving & Storage

- Finished product must be stored and handled under conditions to prevent deterioration.
- Ensure pallets and materials are kept at least 18" away from the walls.
- Inspect torn bags and boxes and repair if appropriate.
- Brush off bags and boxes before opening them.
- Store ingredients and products at the appropriate temperature.
- Ensure stock rotation of ingredients. Use oldest stock first.

- Handle ingredients or products carefully to avoid spilling.
- Do not return products or ingredients to the production line after they have touched the floor or any other surface that is not clean.
- Returned defective product must be clearly identified and isolated in a designated area for appropriate disposition.

Retail Store

- Monitor and maintain proper temperatures.
- Rotate ingredients using first in-first-out (FIFO) and check for expired items.
- Check best before dates and the quality of the food before using.
- Refrigerate cold foods immediately upon receipt.
- Sanitize equipment, cutting boards, work surfaces and utensils.
- Always wash hands after handling money.

Transport

- Food should be transported at the appropriate temperature
- Keep vehicles/containers for transporting products clean, in good repair and condition.
- Check vehicle for cleanliness, chemical spills, foreign objects, and pest infestation before loading.
- Cleaning and sanitizing should take place between loads.
- Food carriers should be inspected by the manufacturers on receipt or prior to loading to ensure they are free from contamination and suitable for the transportation of foods.
- Food carriers should be loaded, arranged, and unloaded in a manner that prevents damage and contamination of the food and packaging materials.
- It is preferred that food carriers carry only food, to avoid cross contamination.
- Incoming materials should be received in an area separate from processing areas.
- Where appropriate, materials used in carrier construction should be verified to be suitable for food contact.

Module 4: Concepts for Food Safety Management in the Food Industry

Food Safety Management across the Supply Chain



Control and minimize hazards from production, across all the steps to the consumer.

This approach is called **"farm to fork"**

Food safety is ensured through the combined efforts of all the parties participating in the food chain.

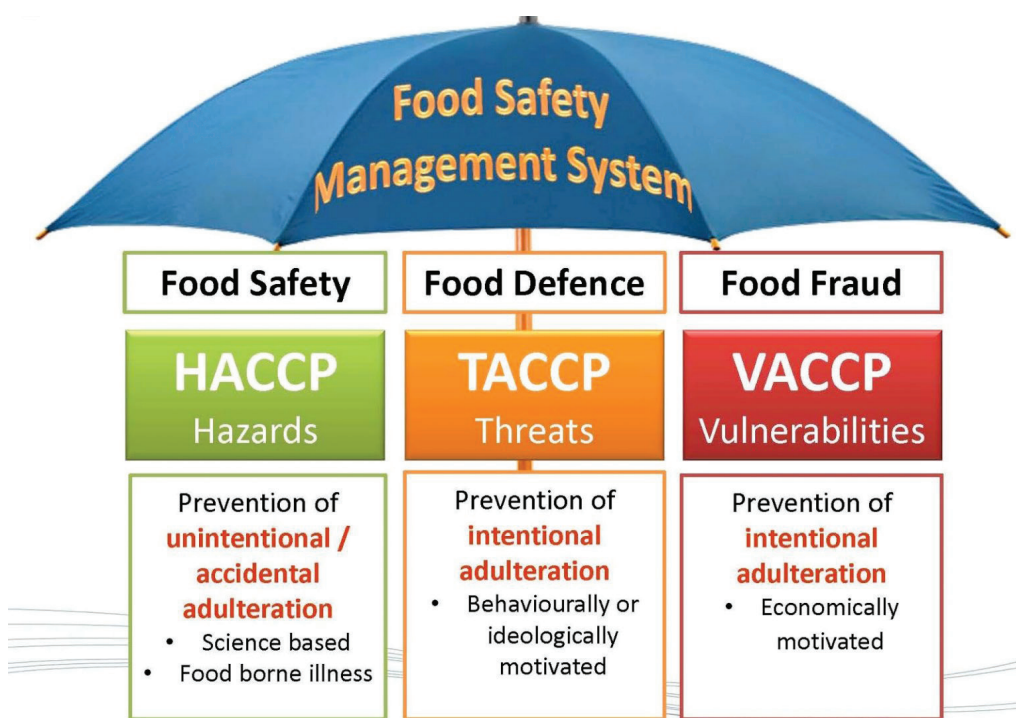


Food Safety Concepts in The Food Industry

The three concepts for maintaining food safety in the food industry include:

- Hazard Analysis and Critical Control Points (HACCP): This addresses the unintentional/ accidental contamination of food during processing operations by food hazards that can lead to foodborne illnesses.
- Vulnerability Assessment and Critical Control Points (VACCP): It addresses food fraud and intentional adulteration of food for economic gain as a mitigation measure.
- Threat Assessment and Critical Control Point (TACCP): This addresses food defence to mitigate against threats to the food industry such as intentional adulteration motivated by ideological reasons to cause harm to the body.

Food Safety Management System by the Global Food Safety Initiative (GFSI)



Hazard Analysis and Critical Control Points (HACCP)

It is a science-based and systematic approach to hazard identification, assessment, and control. Potential hazards (biological, chemical, and physical which may be associated with food from growth, harvesting, processing, storage, and distribution to consumers) are identified to minimize the food safety risk. HACCP is a proactive approach in a food management system to identify potential hazards and determine the right plant equipment to be installed to ensure the safety of food products. The system is flexible to changes as new control measures can be implemented in new equipment design and most importantly helps processors channel resources rightfully. HACCP covers production from farm to fork. It is a measure against food safety hazard.

5 Steps in Hazard Analysis and Critical Control Points (HACCP)

1. Assemble HACCP team: A multidisciplinary team from the company department such as: production, engineering/maintenance, quality control and assurance, food safety, human resources, marketing environment, health and safety, security, and procurement along with top management must be part of the food safety team. Team members must have knowledge of food ingredients and processing, potential hazards during processing operations, equipment design and maintenance, procurement, security surveillance for food defence, storage, and distribution. It is key for them to be able to evaluate hazards, identify controls and limits for effective monitoring and documentation.

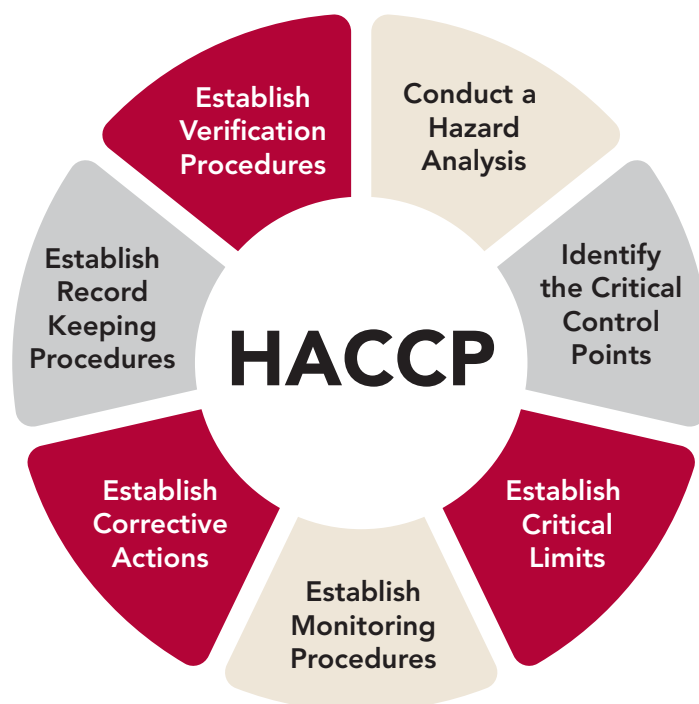
2. **Describe the product:** The food product formulation information such as ingredients to be used, production/processing method, handling, storage, and distribution should be described fully in the HACCP plan to determine potential hazards that can be found in the food product.

3. **Identify Intended use:** The intended use of a food product and target consumers must be clear and written with the type of package, shelf-life, and temperature of storage. Preparation and use must also be stated.

4. **Develop a flow diagram:** A flow diagram provides a clear and simple outline of the steps involved in the production process. Control measures and critical limits are stated clearly in the schematic diagram for easy identification and implementation during the production process.

5. **Verify the flow diagram:** Onsite verification of the flow diagram should be done by the Food Safety team to verify the effectiveness of the HACCP plan and apply modifications to update the flow diagram and process. All steps in HACCP must be clearly documented as references in implementation and for regulatory purposes.

Principles of Hazard Analysis and Critical Control Points (HACCP) Implementation



7 Principles of Hazard Analysis and Critical Control Points (HACCP) Implementation

1. **Conduct a hazard analysis:** Hazard analysis must be conducted by the Food Safety/HACCP team to identify appropriate control measures for food hazards. Hazard analysis is key to an effective HACCP plan and focuses on raw materials and ingredients, steps in processing, product storage and distribution, preparation, and consumption. It involves brainstorming and review sessions to evaluate hazards

that can be found in the production process with potential hazards identified. Risk assessment is done to know the severity of the potential hazard and adequate control measures to prevent the food safety hazard occurrence.

2. Determine Critical Control Points (CCP): CCP is a step in which a control can be applied to effectively prevent or eliminate a food safety hazard or reduce it to an acceptable level. It addresses food hazards that control measures are not effective to eliminate. A decision tree is used to determine CCPs in a food process.

3. Establish Critical Limits: Critical limit is the maximum or minimum value to which a biological, chemical, or physical hazard must be controlled at a CCP to prevent, eliminate, or reduce to an acceptable level the occurrence of a food safety hazard. Each control measure at a CCP has a critical limit which must be scientifically proven. This is based on factors such as temperature, time, physical dimension, humidity, moisture level, water activity (aw), pH, titratable acidity, salt concentration, available chlorine, viscosity, preservative, or sensory information such as flavour, aroma, and visual appearance.

4. Establishment of monitoring procedures: It is planned sequences of observations or measurement to access whether a CCP is under control or deviated to produce accurate record for future verification. Monitoring is important for tracking of processing operations in food safety management system. When deviation occurs, corrective action must be taken. Equipment also need to be properly calibrated and only trained personnel in quality control should be involved in monitoring and documentation.

5. Establish corrective action: Corrective actions are done after deviation at Critical Control Points (CCPs) with critical limits exceeded to prevent foods that may be hazardous to health from reaching consumers. The cause of non-compliance is established, and non-conforming products are disposed of if they will lead to a food safety risk or can be reprocessed if they can be eliminated or reduced to an acceptable level. A corrective action plan must always be documented and kept.

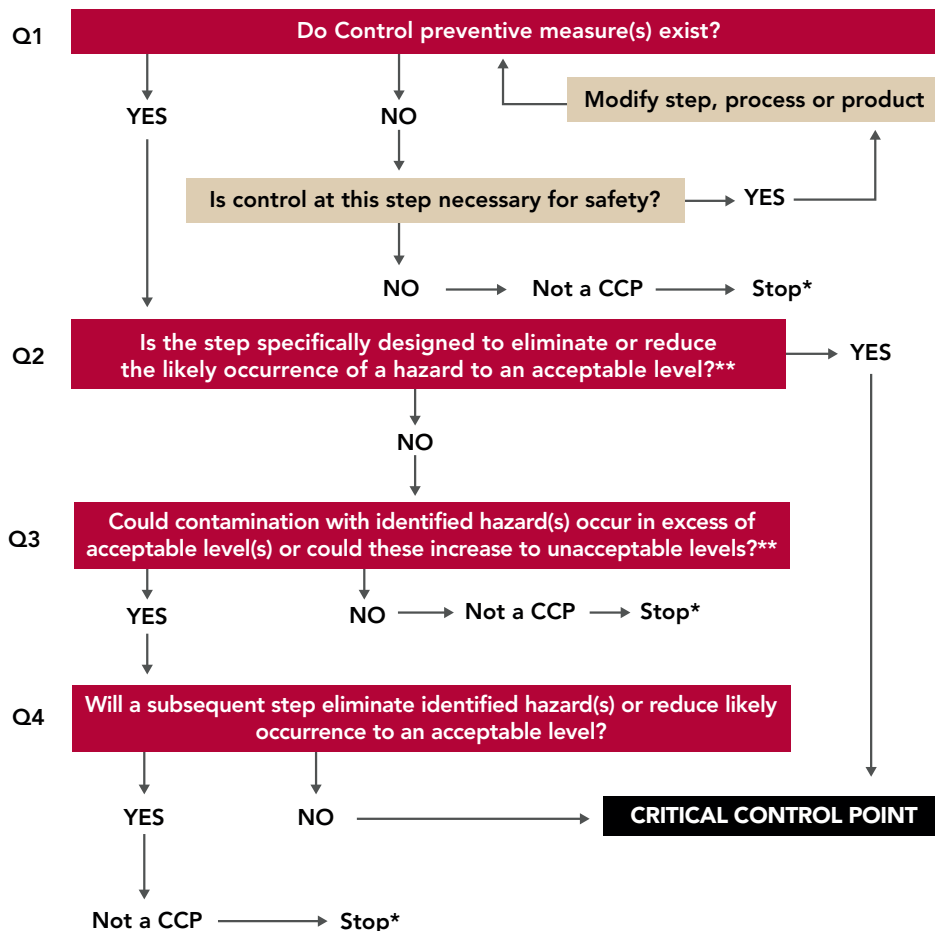
6. Establish verification procedures: It is activities other than monitoring that determine the validity of the HACCP plan. Verification ensures the HACCP plan is being followed with a review of CCP monitoring and corrective action.

7. Establish record keeping and documentation procedure: Records kept should include hazard analysis, food safety team and responsibilities, product description and intended use, flow diagram and verification, monitoring frequency and records, corrective actions, verification procedures, and schedules.

12 Implementation Steps of HACCP

- 1 • Assemble the HACCP team
- 2 • Describe the product
- 3 • Identify intended use
- 4 • Construct a flow diagram
- 5 • On-Site confirmation of flow diagram
- 6 • Principle 1: List all potential hazards associated with each step, conduct a hazard analysis and consider any measures to control identified hazards
- 7 • Principle 2: Determine critical control points
- 8 • Principle 3: Establish critical limits for each critical control point
- 9 • Principle 4: Establish a monitoring system for each critical control point
- 10 • Principle 5: Establish corrective actions
- 11 • Principle 6: Establish verification procedures
- 12 • Principle 7: Establish documentation and record keeping

Hazard Analysis and Critical Control Points (HACCP) Decision Tree



Vulnerability Assessment and Critical Control Points (VACCP)

This addresses food fraud which is an intentional activity for the purpose of economic gain or financial benefit which affects food quality and poses risk to health. It is also referred to as an illegal deception for economic gain. Intentional adulteration poses threat to the food industry and a high risk for food safety which must be evaluated and addressed by the food safety team to mitigate against intentional adulteration.

Types of Food Fraud

Adulteration: It is an intentional act of substituting or adding a substance into a food product such melamine in milk, Sudan red colorant to puree, sauces, and spices.

Tampering: Legitimate products and packages are mislabelled such as changes in expiry date or nutritional information.

Over-run and unauthorized production: It is when production is made in excess of the daily target and under-reported.

Counterfeiting: The replication of popular food products and packages with high food safety risks to the consumer.

Theft: When legitimate food products are stolen and passed off as legitimately procured.

Diversion/Gray Market: Distribution of legitimate products outside intended market. Example is smuggling of relief materials for sale in markets.

Simulation: When a product is designed to look like a popular brand but does not have the same quality and food safety assurance.

Threat Assessment and Critical Control Points (TACCP)

Food defence is the set of measures and actions regulated through procedures to ensure the security of the food and supply chain from malicious and ideological motivated attacks leading to disruption of the food supply chain. It is therefore important in the food industry to protect food facilities from intentional food contamination from primary production to the consumer from these potential threats such as toxic chemical substances, biological toxins, physical and nuclear substances. It can be prevented using access control, surveillance cameras and use of scanners.

The methods of intentional food contamination include:

External attack: When raw materials used for food production are contaminated before supply for food processing.

Forceful entry to production facilities: When a person gains access to a production facility forceful to contaminate food.

Secret entry to food processing facility: When a visitor to a processing facility secretly contaminates food products to cause economic harm to the plant and injury to consumers.

Inside collaborator: When an employee is used to sabotage a food processing facility through intentional contamination.

Module 5: Food Quality Control and Assurance in the Food Processing Industry

Food Quality Control and Assurance

Food Quality: Is the attributes of food that are matched with pre-defined standards and influences its value to consumers.

Food Quality Control: Is a quality management tool applied in the manufacturing and distribution of food products with the optimum level of quality fulfilled. Food quality control ensures conformity, wholesomeness and product reliability as stated in the food product standards. Food quality control is a reactive process that focuses on the identification of defects and is a product-based approach to verifying food quality.

Food Quality Assurance: Is a design plan established to ensure food quality, as defined, is maintained with specified limits. Quality assurance is a proactive approach that focuses on the prevention of defects and is process based to manage food quality.

Food Quality

- Food quality is “a complex characteristic of foods that determines its value and acceptability by consumers.
- A good quality product complies with the requirements specified by the client.
- These may include purity, size, flavour, aroma, colour, maturity, safety, wholesomeness, nutrition, or any other attribute or characteristic of the product.
- Food quality represents the totality of all properties and attributes of a food product that are acceptable to the customer.
- Food safety & food quality are part of important buyers’ requirements in addition to others such as plant / animal health, packaging, and labelling.

Food Quality Control

- A quality control system should be established to ensure that products contain the correct materials of specified quality and quantity and are manufactured under proper conditions following standard procedures to ensure the quality and safety of the product.
- Quality control involves sampling, inspecting, and testing of starting materials, in process, intermediate, bulk and finished products. It also includes where applicable, review of batch documentation, sample retention program, stability studies, product complaints, product recalls, and maintaining correct specifications of materials and products.

Food Quality Assurance

Quality assurance is the part of quality management focusing on increasing the ability to fulfil quality requirements, according to current ISO standards.

The objectives of quality assurance are achieved when processes have been defined which, when followed, will yield a product that complies with its specification and the quality expected, and when the finished product:

- contains the correct ingredients in the correct proportions.
- has been correctly processed, according to the defined procedures.
- is of the purity required.
- is enclosed in its proper container, which bears the correct label (or is otherwise suitably marked or identified).
- is stored, distributed and recommendations given for its subsequent handling in accordance with the recommended storage conditions, so that its quality is maintained throughout its designated or expected life/

Properties of Food Quality

The four properties of food quality are:

Organoleptic quality: It is the use of sense organs for sensory evaluation to determine quality.

Nutritional quality: The use of chemical constituents such as protein content, carbohydrate, moisture content, fat, vitamins, and minerals to determine the quality of a food product.

Functional quality: Attributes such as bulk density, viscosity, swelling, and least gelation capacity are used to determine the quality of a food product as it influences consumer acceptability.

Hygienic quality: These are conditions during processing and handling that may influence the risk of foodborne diseases.

Seven Basic Tools of Food Quality Control

These are used in problem-solving and process improvement in the food industry to enhance food quality.

Cause and effect diagram: It is called the Ishikawa or fishbone diagram which identifies the possible causes for an effect or problem. The generic categories of the cause-and-effect diagram are based on six causes: environment, materials, machine, measurement, man, and method.

Control chart: It is a graph used to show how a production process in the food industry changes over time. It involves comparing current data to historical control limits to know if the process is in control or out of control.

Check sheet: Check sheets are simple forms with certain formats that can aid the user to record data in a firm systematically. They are easy to understand and apply. The three types of check sheets are: defect-location check sheets; tally check sheets, and defect-cause check sheets.

Pareto chart: This is a special type of histogram that can easily be applied to find and prioritize quality problems, conditions, or their causes in a food plant. The aim of Pareto chart is to figure out the different kinds of “nonconformity” from data figures, maintenance data, repair data, parts scrap rates or other sources.

Histogram: It is a type of bar chart that describes frequency distribution and visualizes both attribute and variable data of a product or process, also assists users to show the distribution of data and the amount of variation within a process.

Scattered diagram: It is a powerful tool to draw the distribution of information in two dimensions, which helps to detect and analyse pattern relationships between two quality and compliance variables.

Flow chart: It is a diagrammatic picture that describes the sequence of steps which exist in an operation or process. It visualizes a picture including the inputs, activities, decision points, and outputs for easy implementation to achieve a process objective.

STRENGTHENING NUTRITION IN PRIORITY STAPLES(SNIPS) FOOD SAFETY AND QUALITY GUIDE				
PRIORITY STAPLE CROP				
VALUE CHAIN STAGE	VITAMIN A MAIZE(VAM)	VITAMIN A CASSAVA(VAC)	ORANGE FLESHED SWEET POTATO(OFSP)	RICE
1. Production on the Farm	Good agronomic practices, use of viable seeds, use of herbicides, pesticides and fertilizers in the right dosage.	Good Plant Breeding Practices, Good Agronomic Practices, use of viable cuttings and stems, use of herbicides, pesticides and fertilizers in the right dosage.	Good Plant Breeding Practices, Good Agronomic Practices, use of viable vines, use of herbicides, pesticides and fertilizers in the right dosage.	Good Agronomic Practices, use of viable seeds, use of herbicides, pesticides, and fertilizers in the right dosage.
2. Harvest /Post Harvest Handling	Drying under controlled conditions, Moisture control to atleast 14%, Adequate aeration and temperature control, Mechanical and insect infestation control, Hygienic practices. Store dried maize grains in air tight sack or silos	Ensure the cassava tubers are not injured, process harvested Vitamin A Cassava within 2 days to avoid deterioration. for chips ensure you dry properly while for fufu ensure fermentation is done properly	Harvested carefully and handled properly to avoid bruises. Process OFSP within two days after harvest and store in cool and dry condition(Cover with grasses or store in the ground. Keep away from pest and insect infestation.	Drying under controlled conditions, Moisture control to atleast 14%, Adequate aeration and temperature control, Mechanical and insect infestation control, Hygienic practices, proper storage in sacks or silos
4. Storage and Transportation	Store in airtight sacks properly covered and store under dry condition. Vehicles for transport must be hygienic and covered with a tarpulin to avoid moisture absorption.	Use of grasses to cover the crop away from sunlight. Use of baskets, crates and aerated cartons to transport harvested crops	Use of grasses to cover the crop away from sunlight. Use of baskets, crates and aerated cartons to transport harvested crops	Store in airtight sacks properly covered and store under dry condition. Vehicles for transport must be hygienic and covered with a tarpulin to avoid moisture absorption.
5. Food Safety , Processing and Consumption	Wash properly before cooking/ processing to remove dirt and foreign materials. Check to ensure no greenish or dark spots which may indicate mold contamination. Maize has high rate of aflatoxin contamination which makes it unfit for consumption and important to ensure grain quality. Insect infestation is also a challenge	VAC to be processed to products such as garri and fufu must be fully fermented to eliminate the presence of hydrogen cyanide which is harmful to cardiovascular, respiratory and neurological function in the body. Those to be processed into flour must be dried properly.	Fungal contamination is common with OFSP and it should be stored properly.	High moisture content leads to caking and growth of mold such as <i>Aspergillus flavus</i> and makes rice unfit for consumption. Bacteria contamination is one of the risks associated with rice that has not been appropriately dried and insect infestation such as weevils.
6. Registration with Regulatory Authorities and Commercialization	Ensure processed products from Vitamin A Maize, Vitamin A Cassava, Orange Fleshed Sweet Potato, and Rice are package, properly labelled and registered with National Agency for Food and Drug Administration and Control.			

Module 6: Pre-Requisite Programs (PRP) in Food Processing

Pre-Requisite Programmes (PRP)

PRPs include Good Hygienic Practices, Good Agricultural Practices and Good Manufacturing Practices, as well as other practices and procedures such as training and traceability, that establish the basic environmental and operating conditions that set the foundation for implementation of a HACCP system.

Prerequisite program consists of activities and operations that are directed at maintaining the environment, facilities, structures, and equipment in a food plant under clean and sanitary conditions. It also includes pest control, cleaning and sanitation, waste management, fleet management and personal hygiene. Maintenance of clean and sanitary conditions in a food plant (clean and sanitary environment, facilities, structures, and equipment) is essential to ensure that food is produced under hygienic conditions, to prevent contamination and to prevent breeding of pests.

Good Manufacturing Practices in Processing Operations

GMP deals with contamination arising from people, products (food materials), packaging materials, hazardous materials, and miscellaneous materials.

GMP is a combination of manufacturing and quality control procedures designed to ensure that products are consistently manufactured to their specifications. The application of Good Manufacturing Practices also requires reassessing the potential risk of each food hazard in food processing. Although less severe hazards or hazards not very likely to occur will probably not need to be analyzed in an HACCP plan, they must be addressed within the framework of GMP. The key factors considered are:

Premises: The location of a food plant has effect on its quality and should be in areas to avoid cross-contamination and enable adequate cleaning and maintenance. Construction and layout of premises, roadways, drainages, sanitary facilities, water quality and treatment are very important to maintain GMP.

Equipment: The design of equipment and installation to enable maintenance, calibration, and easy cleaning practices. Materials used in equipment fabrication should be corrosion resistant to prevent contamination of food.

Hygiene, health requirement and training: Staff must maintain personal hygiene to avoid cross contamination of food.

Pest control and sanitation: Food contact surfaces should be non-absorbent, non-toxic, smooth, free from pitting, unaffected by food and able to withstand repeated cleaning and sanitization. Pest control must be carried out by trained personnel or consultants and should not contaminate food products.

Record keeping and documentation: Cleaning, sanitation and pest control records must be kept including the schedule and frequency.

The 5P's of Good Manufacturing Practices



Siting of a Food Establishment

Establishments should normally be sited away from: Environmentally polluted areas and industrial activities such as dump sites, quarries, abattoirs, sewage treatment sites, sawmills which pose a serious threat of contaminating food. Potential sources of contamination need to be considered when deciding where to locate food establishments, as well as the effectiveness of any reasonable measures that might be taken to protect food.

In plant location and design, these factors must be considered:

- Location of food establishments can be a potential source of contamination as obnoxious odour, fumes, excessive soot, dust, smoke, chemical or biological emissions and pollutants from surroundings can enter food.
- Internal design, structures and layout of the premises rooms and equipment should be such that they prevent contamination.
- Material of equipment should be durable, movable, and capable of being disassembled for cleaning.
- Premises and equipment should be easy to maintain, clean, disinfect and monitor.
- Areas that are subject to flooding unless adequate safeguards can be provided
- Areas prone to infestations and pests should be avoided.
- Areas where wastes (solid, liquid) cannot be effectively removed are not suitable.

Location of Equipment

Equipment should be located such that it:

Permits adequate and effective maintenance and cleaning: The layout and the location of all structures, equipment, services, and facilities should provide sufficient space for movement of personnel and equipment, flow of air, materials and products, movement of waste and garbage, and storage of materials and products, to prevent contamination and cross-contamination of materials and products.

Functions in accordance with its intended use: There should be physically separate areas that are designated for processing, for packaging, for raw materials storage and preparation, and for finished product storage to avoid cross-contamination. Facilities for personnel, laboratory and other service areas should be included in the design with proper specifications.

Facilitates good hygiene practices and monitoring: Activities that could become potential sources of cross-contamination should be located in areas that are separated from other processing activities and movement between high and low risk areas of the factory is minimized with controlled corridors.

Equipment and Utensils

- All equipment, utensils and packaging materials coming in contact with food must be cleanable and capable of being disinfected so as to prevent contamination of the food.
- Equipment and utensils should be made of food grade and durable materials.
- Food control and monitoring equipment should have the capacity to monitor critical limits and maintain conditions required to ensure food safety and its suitability is achieved such as temperatures, humidity, and acidity of food.
- Containers for storage of waste and non-food substances should be identifiable, suitably constructed and, where appropriate, made of impervious material.
- Containers used to hold dangerous substances should be identified and, where appropriate, be lockable to prevent malicious or accidental contamination of food.

Building And Processing Areas.

Design and layout of food plants: It should permit adequate location of equipment and machinery to facilitate maintenance and cleaning. It should permit good hygiene practices including protection against cross contamination between and during food operations.

Internal structures and fittings: It should be built of sound and durable materials that can be maintained easily, cleaned, and disinfected. Floors should be made of smooth, non-absorbent materials, and inert to the food, the detergents and disinfectants under normal operating conditions. The use of wooden food-contact surfaces is not recommended, as they are potential contaminants and are a good harbourage of microbial contamination as it is impossible to sanitize wooden floors.

Temporary/mobile premises or vending machines: It should be well suited, designed and constructed to avoid as far as possible potential food safety hazards and harbouring pest. Food safety hazard associated with these facilities should be adequately controlled to ensure the safety and suitability of food.

Water Supply, Drainage, Cleaning, Lightening, and Storage Facilities

Water supply: There should be adequate supply of potable water with appropriate facilities for its storage, distribution, and temperature control. Non-potable water if used, for example for fire control, steam production and other similar purposes should not contaminate food.

Drainage, Waste Disposal, and Cleaning: Systems should be provided so that the risk of contaminating food or water supply is avoided with areas for cleaning food materials, utensils, and equipment suitably designated. Cleaning facilities should have an adequate supply of hot and cold potable water where appropriate. Non-potable water systems should be identified and not connected with, or allow to reflux into, potable water systems.

Lighting: Light intensity should be adequate and where appropriate; lighting should be protected to ensure that food is not contaminated by breakages.

Storage: Facilities for the storage of food, ingredients, and non-food chemicals (e.g., cleaning materials, lubricants, fuels) should be provided.

Food storage facilities should be designed and constructed to:

- Permit adequate maintenance and cleaning.
- Avoid pest access and harbourage.
- Enable food to be effectively protected from contamination during storage.
- Provide an environment which minimizes the deterioration of food.
- Provide separate storage facilities for cleaning materials and hazardous substances.

Cleaning Procedure

Cleaning methods such as Cleaning in Place (CIP) and Cleaning Out of Place (COP) can be deployed during before and after processing. Cleaning procedures involve:

- Removing gross debris from surfaces.
- Applying a detergent to loosen soil and bacterial film and hold them in solution or suspension.
- Rinsing with clean potable water to remove loosened soil and residues of detergent.
- Dry cleaning or other appropriate methods to remove and collect residues and debris.
- Where necessary, disinfecting with subsequent rinsing except manufacturer's instruction says otherwise.

- Cleaning by the separate or combined use of physical methods, such as heat, scrubbing, turbulent flow, vacuum cleaning, etc

Cleaning Program

- A regular cleaning and disinfection schedule should be drawn up for each establishment to ensure that all areas are appropriately cleaned.
- Cleaning program should be routinely implemented as per the cleaning schedule and should cover the washing tanks, scraping and peeling areas, drying areas and finished product storage.
- Periodical inspections need to be carried out by a supervisor in monitoring the effectiveness of the implementation of cleaning programs.
- Documented cleaning procedures shall be kept and include responsibility for cleaning, item/area to be cleaned, frequency of cleaning, method of cleaning, cleaning chemicals and concentrations, cleaning materials to be used, cleaning records and responsibility for supervision.

Waste Management

- Waste material should be kept separate from work in progress or final products and removed from the workplace daily or more frequently when required.
- It is recommended that each type of waste be clearly marked to show the type of waste that should go into the container.
- Waste bins themselves should be on a cleaning regime to ensure that they do not become the harbourage for bacteria or pests.
- If the waste bin is to have a lid, then it is recommended that this lid is foot operated so that employees do not have to contaminate their hands by lifting the dirty waste bin lid.
- Larger waste bins that are located outside of the building should be fitted with lids to prevent pest activity and should be located away from doors that open regularly to prevent any pest activity becoming an issue within the facility.

Pest Control Program

Pests pose a major threat to the safety and suitability of food and can be prevented by the following conditions:

- Maintain good sanitation, inspection of incoming materials, and good general monitoring.
- Keep buildings in good repair and condition to prevent pest access.
- Prevent accumulation of food residues and water to prevent pest harbourage and infestation.
- Regularly monitor establishment and surroundings to detect and identify evidence of infestation practices.
- Eradicate pest infestation using various methods: physical, biological, and chemical.

Personal Hygiene

- Personnel should always wash their hands: i). at the start of food handling activities; ii). after break; iii) immediately after using the toilet; iv) after handling raw food or any contaminated material; and v). avoid direct handling of ready-to-eat food.
- Where necessary, to minimize microbiological contamination, employees should use disinfectant.
- Protective clothing, hair covering, footwear and/or gloves appropriate to the operation that the employee is engaged in, should be worn, and maintained in a sanitary manner.
- Food handlers should maintain a high degree of personal cleanliness, wear suitable protective clothing, head covering and footwear.
- Cuts and wounds, where personnel are permitted to continue working, should be covered by suitable waterproof dressings.
- People known, or suspected, to be suffering from, or to be a carrier of a disease or illness likely to be transmitted through food should not be allowed to enter any food handling area if there is a likelihood of their contaminating food.
- Any person so affected should immediately report illness or symptoms of illness to the management.
- Medical examination of a food handler should be carried out when such personnel is under suspicion of suffering from illness.
- The following conditions should be immediately reported to management: jaundice, diarrhoea, vomiting, fever, sore throat with fever, visibly infected skin (boils, cuts) and discharge from the ear, eye or nose.
- People engaged in food handling activities should refrain from behaviour which could result in contamination of food such as smoking, spitting, chewing, and eating, talking; and sneezing or coughing over unprotected food.

Fleet and Transport Management

- Food must be adequately protected during transport with the package not contaminated.
- The type of conveyance or container required depends on the nature of the food and the conditions under which it must be transported.
- Different types of foods must be stored separately in the transport vehicle if used for multiple purpose.
- Ready to eat food should be transported in different vehicles separate from those used to transport live animals.
- Conveyance and containers must be in a state of cleanliness and those used in transporting other foods or non-foods must be cleaned effectively and disinfected.
- Ingredients requiring refrigeration should be transported at 4°C or less and should be appropriately monitored. Frozen ingredients should be transported at temperatures that do not permit thawing.
- Food products should be transported under conditions that prevent microbiological, physical, and chemical hazards.
- Open transport vehicles must be properly covered with tarpaulin to prevent moisture absorption, dust, fumes, and heat.

Training

The following should be considered to determine the type of training program required for a Food Business Operator (FBO):

- The nature of the food being handled, its ability to sustain growth of pathogenic or spoilage microorganisms.
- The way the food is handled and packed, including the probability of contamination.
- The extent and nature of processing or further preparation before final consumption.
- The conditions under which the food will be stored.
- The expected length of time before consumption.
- Food hygiene training is important, and all personnel should be aware of their role and responsibility in protecting food from contamination or deterioration.
- Food handlers should have the necessary knowledge and skills to enable them to handle food hygienically.
- Workers that handle strong cleaning chemicals or other potentially hazardous chemicals should be instructed in safe handling techniques.
- Training should be appropriate to the complexity of manufacturing process and the task assigned.
- Personnel should be trained to understand:
 - i. The importance of the critical control points for which they are responsible.
 - ii. The critical limits and the procedures for monitoring.
 - iii. The action to be taken if the limits are not met and the records to be kept.
- Personnel responsible for maintenance of equipment having an impact on food safety should be appropriately trained to identify deficiencies that could affect product safety and to take appropriate corrective action, e.g., in-house repairs, contract repairs.
- Individuals performing maintenance on specific equipment, e.g., closing machine, recorders, etc., should be appropriately trained.

Module 7: Food Traceability

Food Traceability

Food traceability is the process of tracking a food product or raw material in the chain of production.

There are two types of traceability:

Forward traceability: It is the forward track and trace of a food product that has been manufactured but discovered to have a defect which needs to be recalled. The batch /lot number is used to trace the product from the warehouse to the chain of distribution to recall them for management/regulatory decision. Most often forward traceability is because of customer complaint.

Backward traceability: It is the backward track and trace of raw materials used in the manufacture of a food product that has been discovered to be defective through quality control checks. It is used to identify the root cause of a defective product which is isolated to ensure food safety and quality.

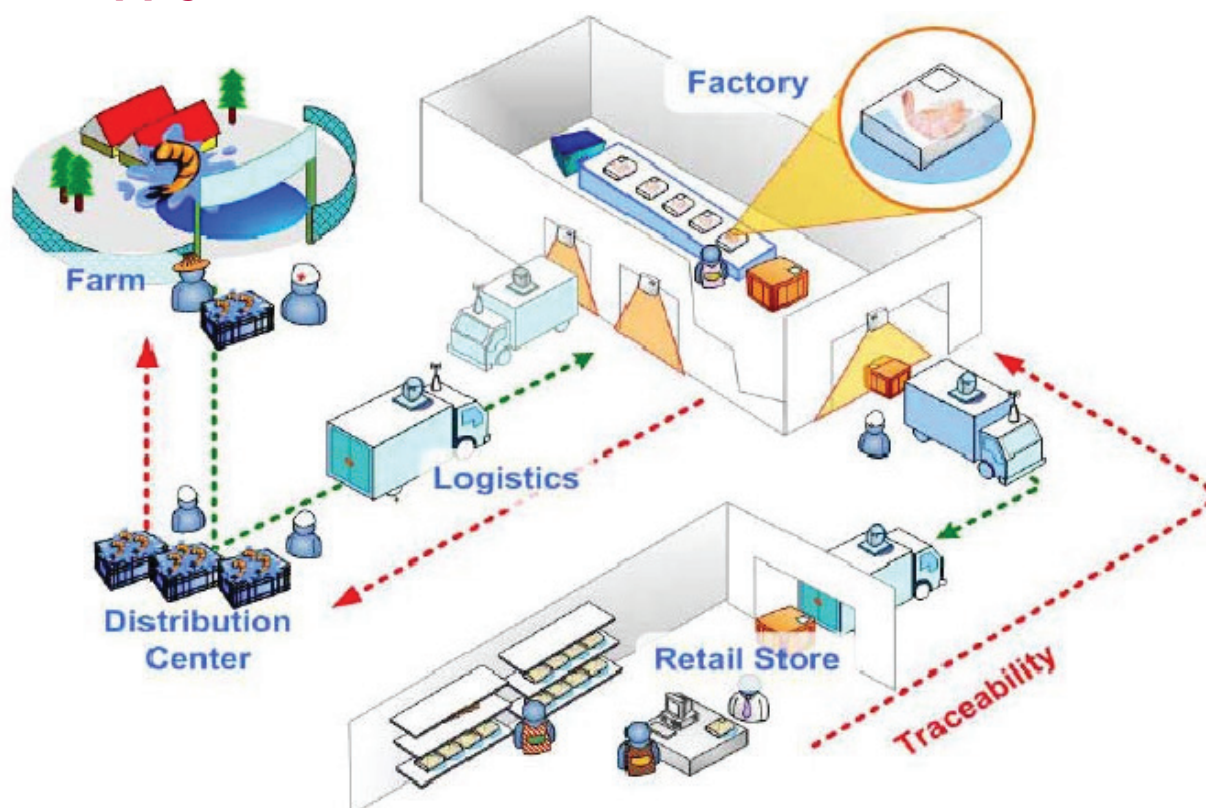
- The ability to identify the origin and history of a product is a critical factor for risk identification and control.
- Each container of food should bear an indelible mark that identifies the producer and lot.
- Traceability makes it possible to trace the food products back to their source (identification of producers, packing operators, etc.).
- The information it provides can be used to isolate and eliminate product lots that present a risk to public health
- Traceability enables producers to withdraw or recall products not in conformity, where necessary.

Important Consideration for Food Traceability

- Receiving, collating, analysing, and reviewing of customer complaints related to defective products ensures the identification of trends and/or an indication of a need for a product recall or of any specific problem requiring attention.
- Rejection of product will be necessary from time to time to: i) protect the public, ii) comply with appropriate legislative or local authority requirements, iii) protect the company or brand name and iv) secure the cost recovery.
- The accurate records of the usage and movement of raw materials, ingredients, and products, including recycled or reworked products, at all stages of processing, handling, storage, and distribution of a food, are essential to trace the correct lots that have been affected.

- A procedure for traceability should be developed to enable the identification of processing history of products produced from the plant at any stage to get details of raw materials, ingredients, processing aids, packaging materials, and the processing and storage equipment used to produce a particular food.
- A written product recall procedure should be in-place to provide the precise method of notifying all distribution networks and retailers involved, including contact names and telephone numbers, and must include halting the transit of products at any stage in the distribution chain.
- All recalled materials should be quarantined, clearly labelled, and held pending when a decision is made by management and regulatory agencies.

Food Supply Chain



Role of Managers and Supervisors in Food Traceability

Managers and supervisors should:

- Ensure effective procedures are in place to control any identified food safety hazard.
- Complete, rapid recall of any implicated batch of the finished food from the market.
- Where a product has been withdrawn because of a health hazard, other products produced under similar conditions should be evaluated for safety.
- Recalled products should be held under supervision until they are destroyed or reprocessed in a manner that ensures their safety.

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